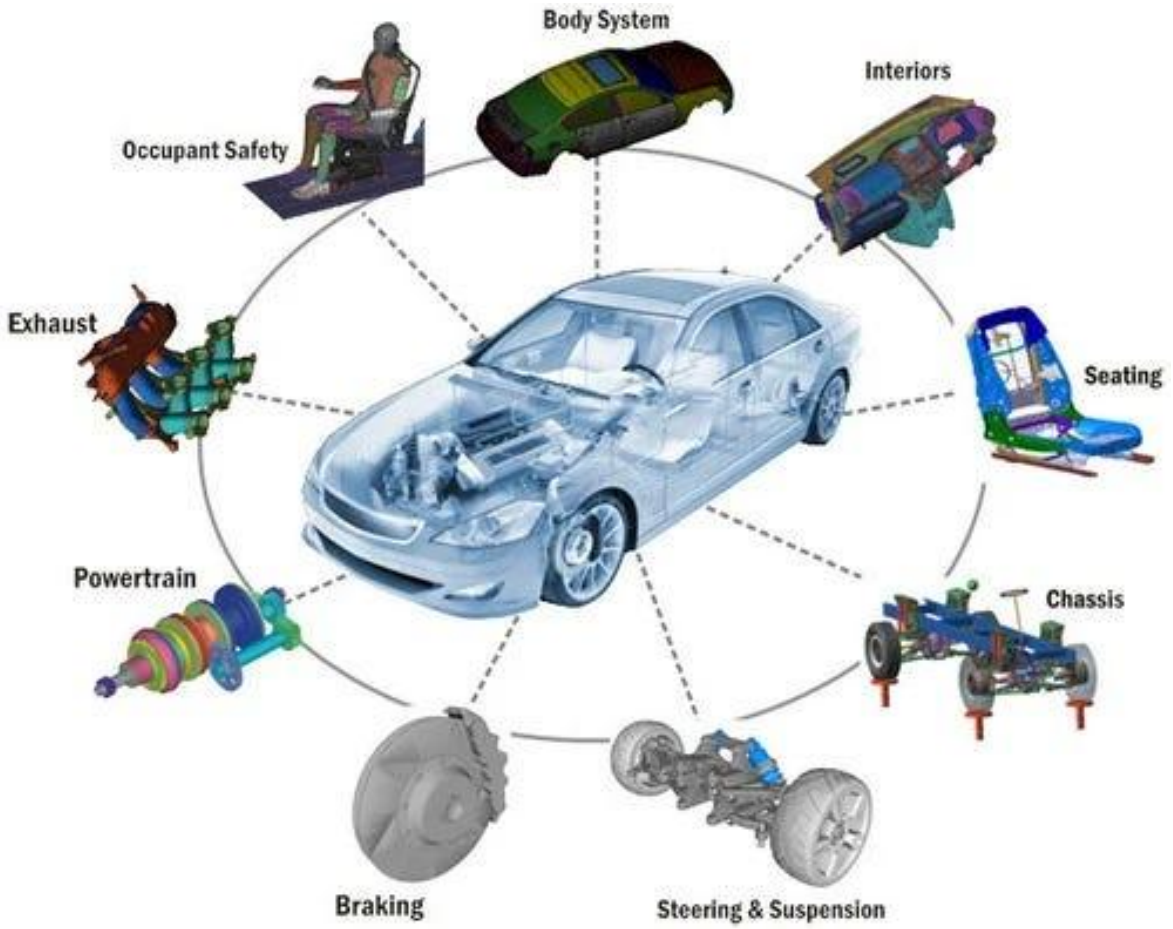
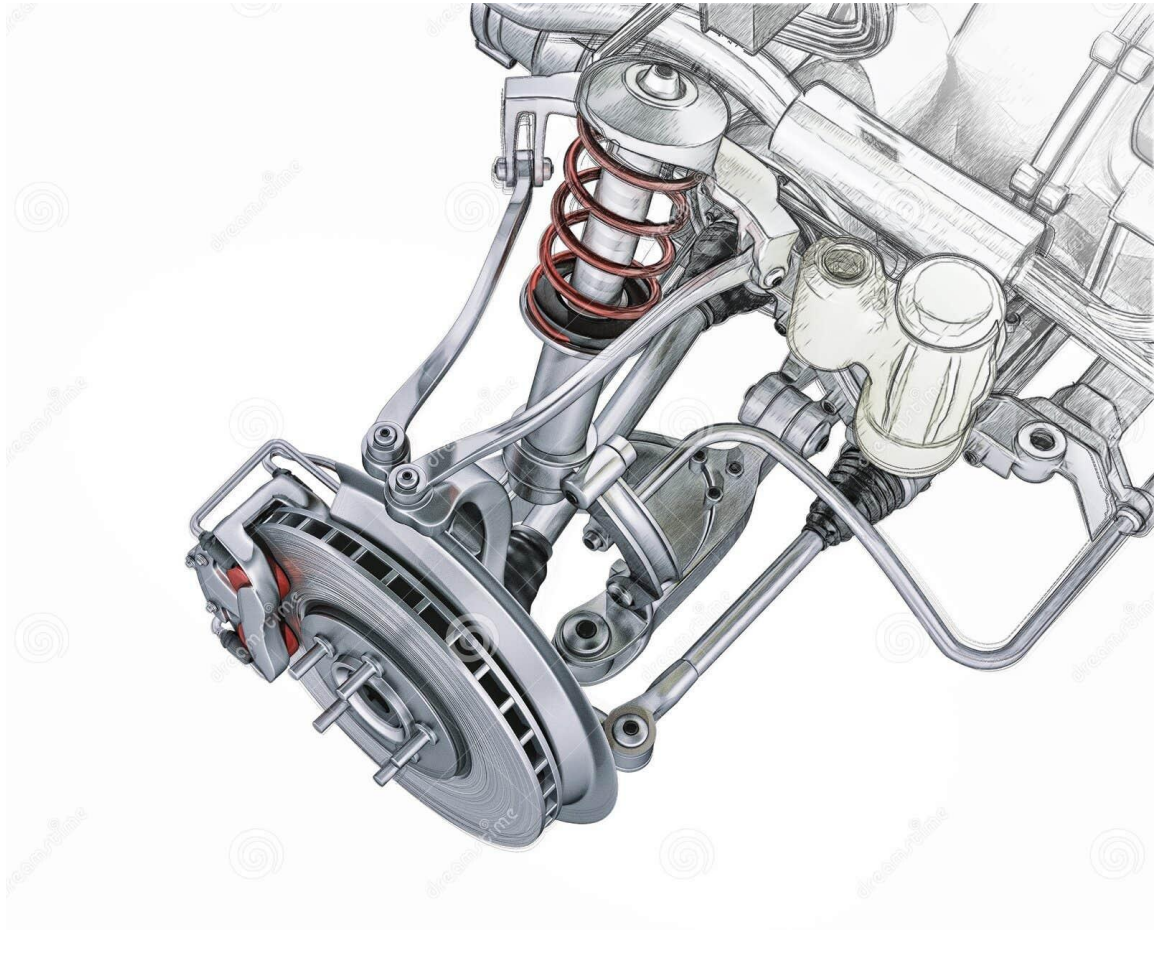




Automotive Engineering (MAM 336)

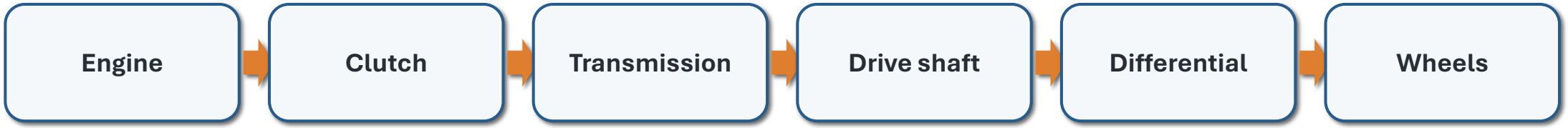
Week 4

Gear Box

DR/ MOHAMED EL TAHER MOHAMED

Power Flow Through the Vehicle

The drive train indicates the power flow from the engine to the driving wheels.



Power source

Torque / speed conversion

Torque distribution

Introduction

The road conditions are not uniform whenever we are in travel in the vehicle. Moreover we do not drive the vehicle always at the same speed. The vehicle must be operated at lower speed when it needs more pulling power. Moreover based on the operating conditions, the vehicle requires different speed and torque. To perform this, a box containing different gears, shafts and bearings is required. This box is called as the gear box. It can be also called as teeth wheel box. By using this box, the driver can achieve different speeds and torques for a vehicle. Gear box is the one which comprises of various teethed gears for different vehicle speed conditions.

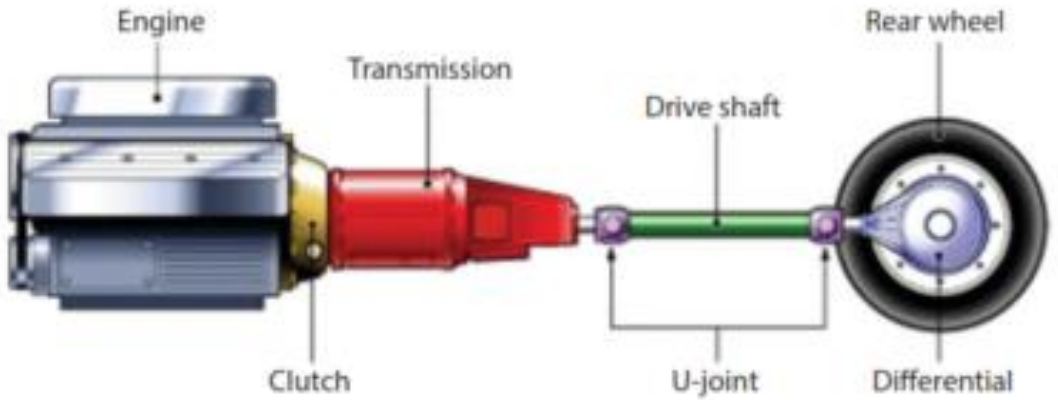


Figure 1.7 Transmission system.

Gear box is located at the second place of the vehicle's transmission system. In the vehicle after the engine, clutch is located. After the clutch gear box is located. It is located in between the clutch and the propeller shaft.

Types of gears

Gears are of many different types. The following are some of the important gears

1. Spur gear
2. Helical gear
3. Double helical gear
4. Bevel gear
5. Rack and pinion gear
6. Worm and worm gear
7. Internal gear

Types of gears

It is like a circular shaped one made on a cylindrical metal having teeth cut in parallel to the axis of the cylinder. It is called as spur gear.



Figure 3.2.1.1 Spur gear

This gear is used to transmit the rotary power only between two parallel shafts. It is capable of transmitting moderate force and speed. A smaller gear is called as pinion. Normally in gear box this type of gear is used for first and reverse gear speed operations.

Types of gears

It is also a circular shaped metal made on a cylindrical metal having the teeth cut not parallel to the axis but in helical shaped cut at certain angle inclined to the cylinder's axis. This type of gear is called as the helical gear.



Figure 3.2.1.2 Helical gear

We can transmit the rotational power to two shafts which are either parallel or non parallel through the helical gear. Compared to straight spur gears more power can be transmitted through helical gears. This type of gear is used in synchromesh gear box and constant mesh gear box.

Types of gears

In this gear the teeth are cut on the top surface of the shaft in tapper. This type of gear is generally used to connect the shafts which are perpendicular to each other and transmit the power in the perpendicular i.e. at 90° angle direction. Bevel gear is capable of withstanding more power than other gears. The bevel gear is used in differential, pinion, crown, wheel, planet and sun gears.



Figure 3.2.1.3 Bevel gear

Types of gears

In this arrangement teeth are cut on a long flat metal bar called as rack. A small wheel called as pinion is in mesh with rack. In this way the rotating power of the pinion is transformed into a straight line or reciprocating motion. This type of gears are used in the gear boxes in the steering system and in many mechanical machines.



Figure 3.2.1.4 Rack and pinion gear

Types of gears

It is used to transmit the rotational power by connecting two shafts that are perpendicular to each other. The gear which looks like a spur gear is called as the worm. The small shaft with the single start thread or multi start thread cut made on it is called the worm shaft. This type of gear is used in places where high speed reduction and high power transmission are obtained. It is used in steering gear box in automobiles.



Figure 3.2.1.5 Worm and worm gear

Types of gears

The internal gear is the one in which the teeth are cut internally on the inner circular metal pipe. This is called as the internal gear. This gear could transmit more power in short places. They are strong in structure. They are used in a small mechanisms and tractor mechanism.



Figure 3.2.1.7 Internal gear

Needs of gear box

1. For reducing the engine speed and increasing the torque.
2. For reducing the engine torque and increasing speed.
3. For the selection of speed levels according to the driver's requirement.
4. For pushing the vehicle in the backward direction.
5. For neutralizing the vehicles when there is no need to transmit the engine power.

Types of gear box

1. Sliding mesh gear box
2. Constant mesh gear box
3. Synchromesh gear box
4. Epicyclic gear box
5. Automatic gear box

Sliding mesh gear box

Sliding mesh gear box was used in earlier day's vehicles. Figure 3.5.1 presents the details of the sliding mesh gear box. In this gear box spur gears are present. This box is suitable for heavy vehicles. By sliding the gears this type of gearboxes move the gears and mesh them and hence it is called as sliding mesh gear box.

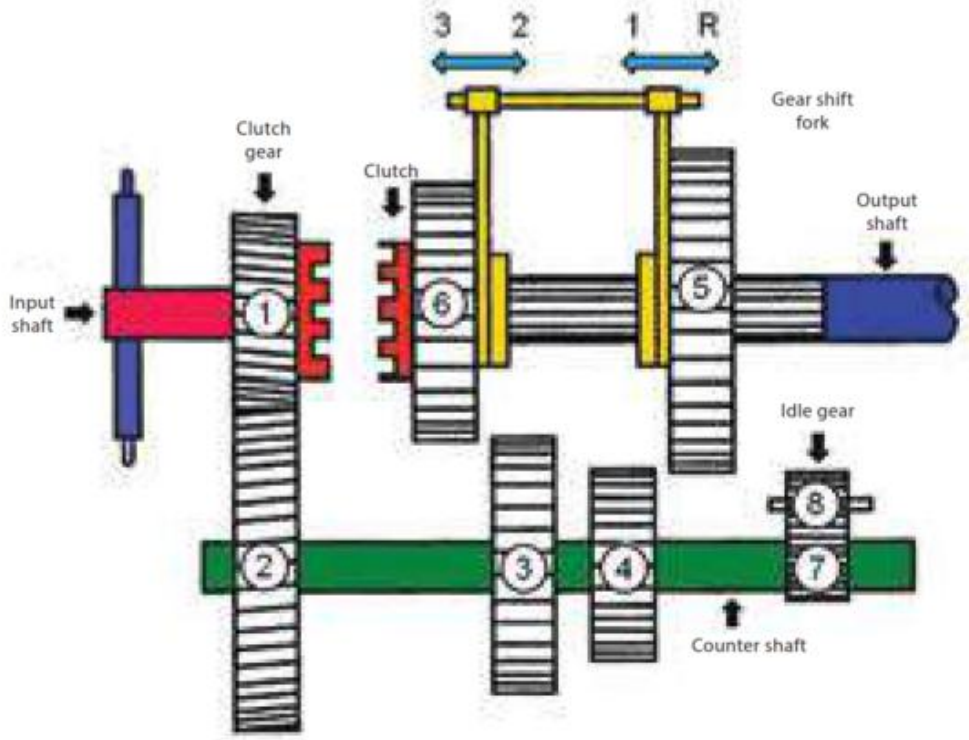


Figure 3.5.1 Sliding mesh gear box

Working principle

a) Neutral position

The gear teeth (1) in the input (clutch) shaft is in mesh with the counter shaft gear 2. Hence the counter shaft rotates. However the gears (3 or 4) are not in mesh with the output shaft gears (5 or 6) as shown in Figure 3.5.1.1. Hence the power will not be transmitted to the output shaft. This condition is called as neutral position.

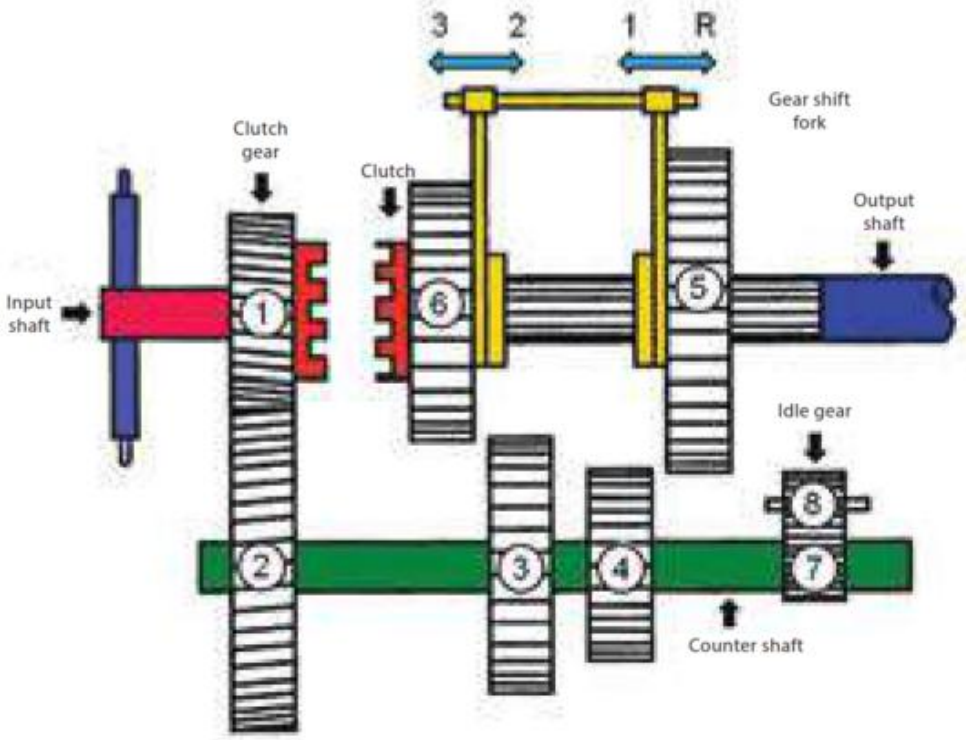


Figure 3.5.1 Sliding mesh gear box

Working principle

b) First gear

The gear (1) in the clutch shaft is in mesh with the gear in the counter shaft (2). As the gears 3 and 4 are in the counter shaft they also rotate. When the gear shift fork is moved towards left as shown in Figure 3.5.1.1 to engage the gear 5 in the output shaft with the gear 4 in the lay shaft, then the output shaft rotates. The rotary power is now transmitted to the output shaft from gear 1-2-4-5. Now we can get first gear ratio 3:1.

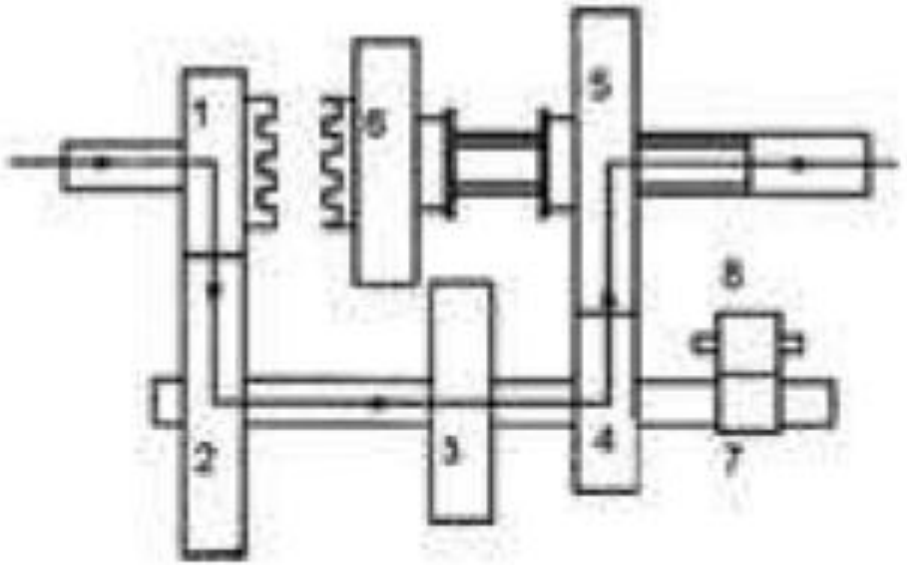


Figure 3.5.1.1 First gear power transmission

Working principle

c) Second gear

During second gear, the gear shift fork is moved right as shown in Figure 3.5.1.2 to mesh the gear 6 in the output shaft with gear 3 in the lay shaft gear 1, 2, 3 and 6 will get contact and rotate. Now the power is transmitted from the gear 1-2-3-6. Now the speed becomes more than the first gear and the gear ratio of 2:1 is achieved.

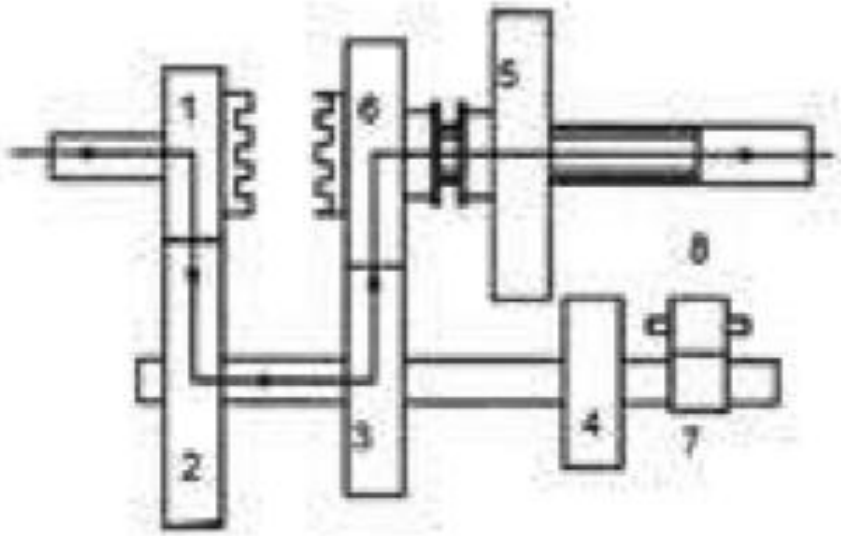


Figure 3.5.1.2 Second gear power transmission

Working principle

d) Third gear

Dog Clutch is located on the gear 1 in the input (clutch) shaft and the gear 6 in the output shaft as shown in

Figure 3.5.1.3. Now by moving the gear shift fork on the left as shown in Figure the dog clutch is allowed to mesh with the dog clutch in the clutch shaft.

Now the output shaft rotates at the speed same as the clutch (main) shaft. This is the maximum speed and the gear ratio in this condition is 1:1. The rotational power is now directly transmitted from the clutch shaft to the output shaft.

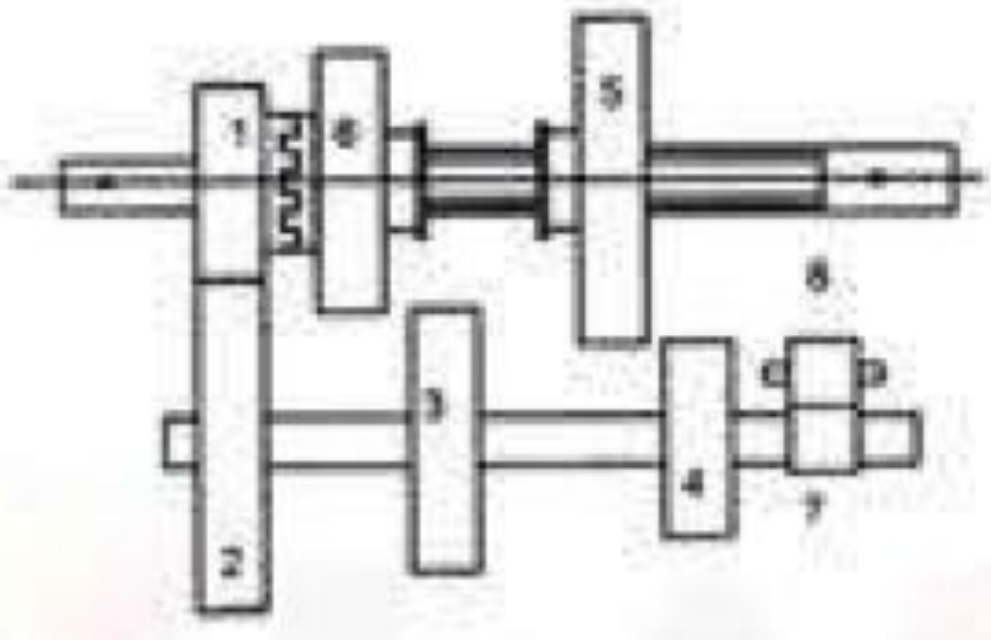


Figure 3.5.1.3 Third gear power transmission

Working principle

e) Reverse gear

For reverse gear, the gear shift fork is moved towards the right direction as shown in the Figure 3.5.1.4. Hence the gear 5 in the output shaft is in mesh with the gear 8 in the lay shaft which is the idler gear. Now the power transmission is from gear 1-2-7-8-5 to the output shaft.

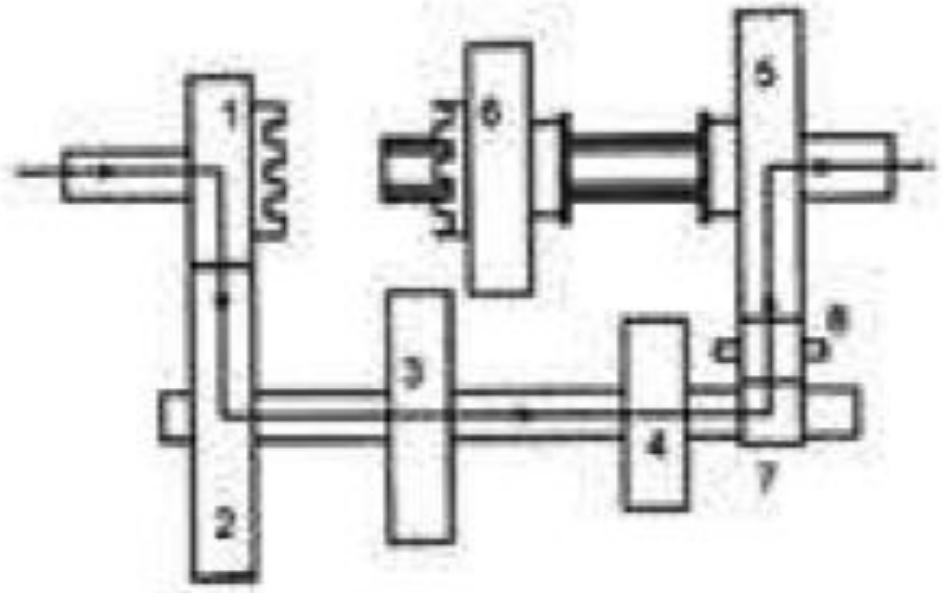


Figure 3.5.1.4 Reverse gear power transmission

Sliding mesh gear box

Advantages of sliding mesh gearbox

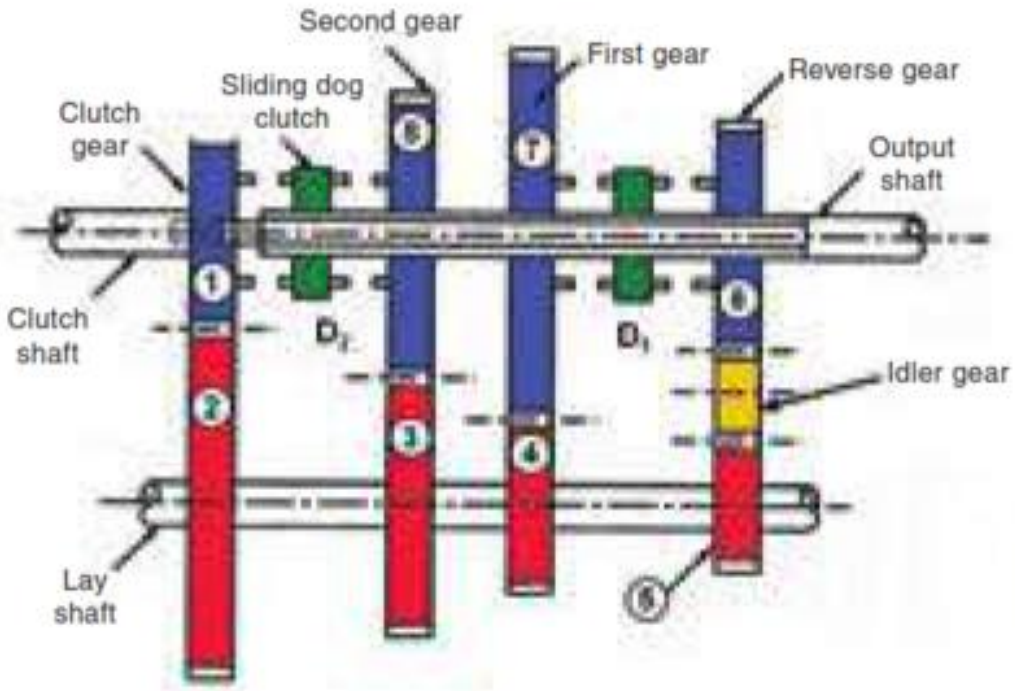
1. Simple in construction
2. Simple in maintenance
3. Suitable for heavy vehicles
4. Low cost
5. Higher mechanical efficiency
6. Production cost is low

Disadvantages

1. Gear changing is difficult
2. Noisy in operation
3. Experience is required for changing gear
4. The edges of teeth wear rapidly.

Constant mesh gear box

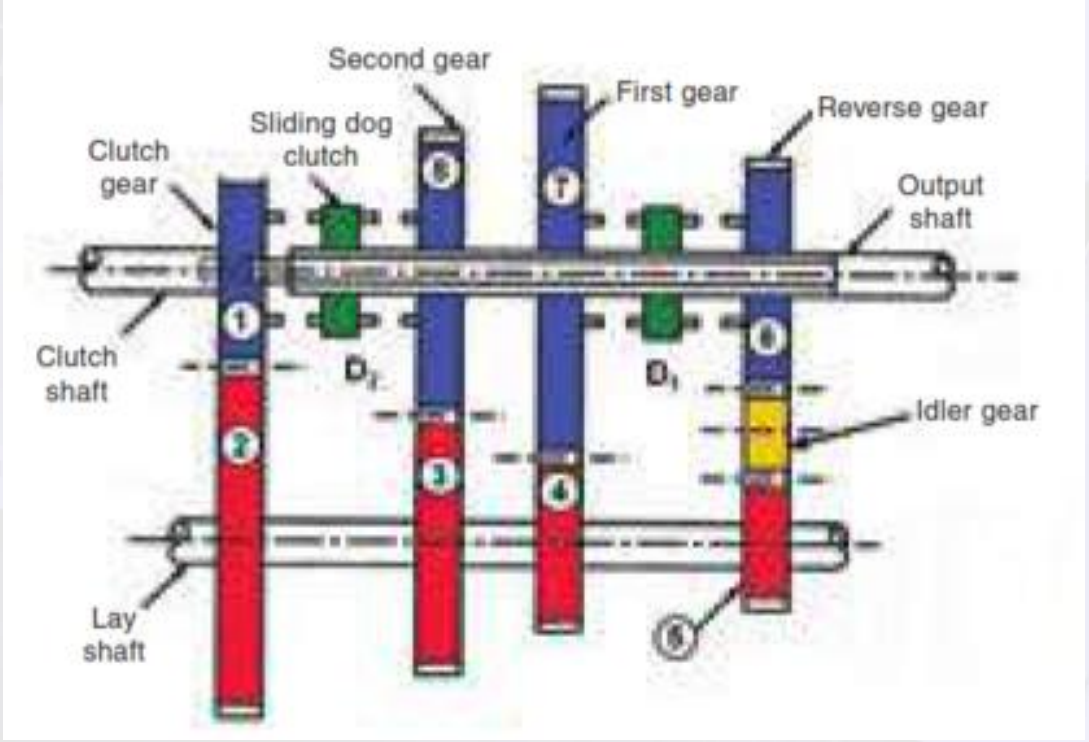
In constant mesh gear box the teeth in the main shaft and lay shaft are always in constant mesh with each other. Hence it is called as constant mesh gear box. Helical gears are used in this type of gear box. The gear arrangements in the constant mesh gear box can be seen in Figure 3.5.2.



Working principle

Neutral position

During neutral condition both the dog clutches (D_1 and D_2) are not in mesh with any gears as shown in Figure. All gears in the shafts 1,2 & 3,4 & 5,6 & 7,8,9 rotate. However the main output shaft does not rotate as the dog clutches are not in engagement with any of the gears. Hence rotational power is not transmitted.



Working principle

First gear

By actuating the gear shift lever, the dog clutch D_1 is moved towards left side to engage gear 7 in the main shaft as shown in Figure. Now the rotational power is transmitted to the main shaft through the dog clutch. Now the speed is low. The power transmission from the clutch shaft to the output shaft is by gear1-2 is means rotational power is transmitted to main shaft by 1-2-4-7-D_1.

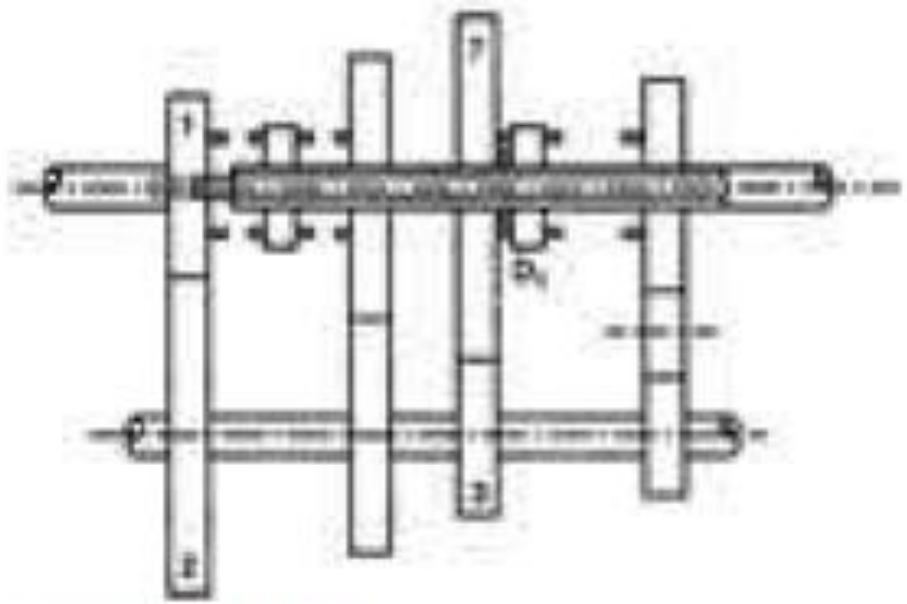


Figure 3.5.2.1 Power transmission in first gear

Working principle

Second gear

For obtaining the second gear, by actuating the gear shift lever, the dog clutch D_2 is moved towards right as shown in Figure to engage the gear 8 in the output main shaft. Now the rotational power is transmitted to the main shaft through the dog clutch D_2. Now the speed of the main shaft is higher than the speed of the first gear. The power transmission from the clutch shaft to the output shaft is through the gears1-2-3-8-D_2.

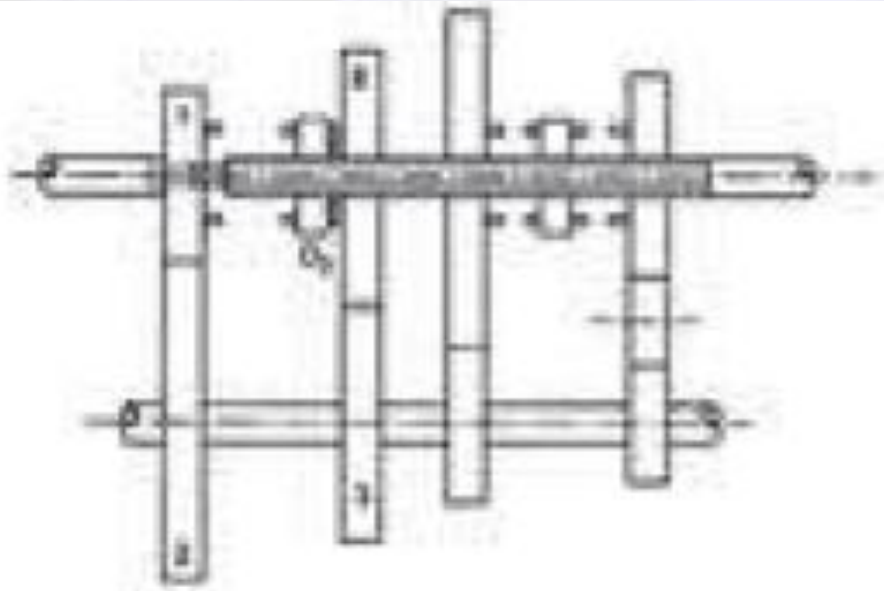


Figure 3.5.2.2 Power transmission in second gear

Working principle

Third gear

To achieve the third gear the dog clutch D_2 is moved left to engage with the clutch shaft gear directly. Now the output shaft rotates at the speed same as the clutch shaft speed. The rotational power is transmitted to the main shaft directly through the gear1-D_2.

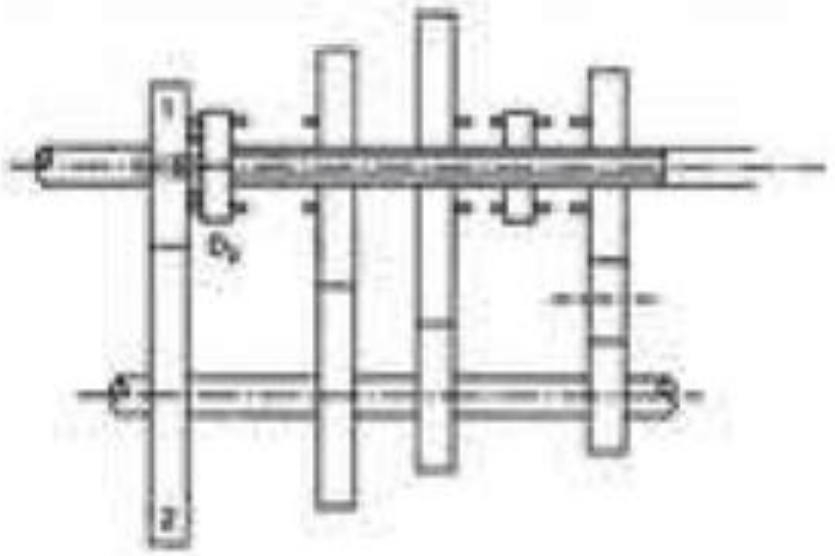


Figure 3.5.2.3 Power transmission in third gear

Working principle

Reverse speed

To obtain the reverse gear, the dog clutch D_1 is actuated to move right towards the reverse gear 6 to engage as shown in Figure 3.5.2.4. As the idler gear is placed between the gears 5 in the lay shaft and 6 in the main shaft the idler gear changes the direction of rotation of the gear 6 in the main shaft. Now the vehicle moves in the reverse direction. The power transmitted from the cluth shaft to the main shaft is through gears 1-2-5-Idler-6-D_1.

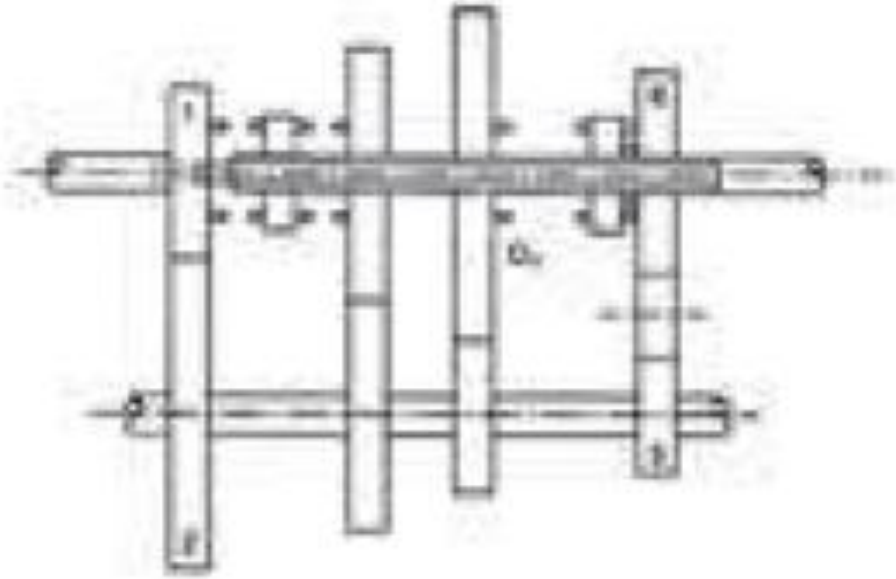


Figure 3.5.2.4 Power transmission in reverse gear

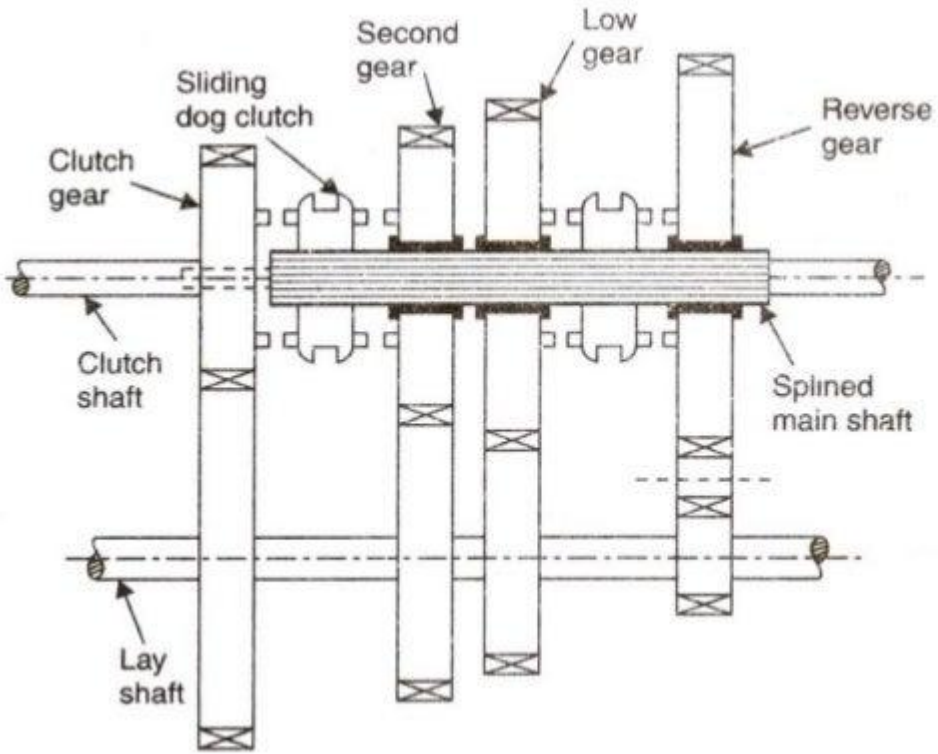
Constant mesh gear box

Advantages of constant mesh gear box

1. Gear changing is simple
2. Less noise
3. Less chance for the teeth of the gears to break up.
4. Smoothness in operation
5. Low maintenance

Disadvantages

1. For changing gears from one speed to other speed, double de clutching has to be done.
2. More wear
3. Possibility of wear in dog clutch.





Thank You!